

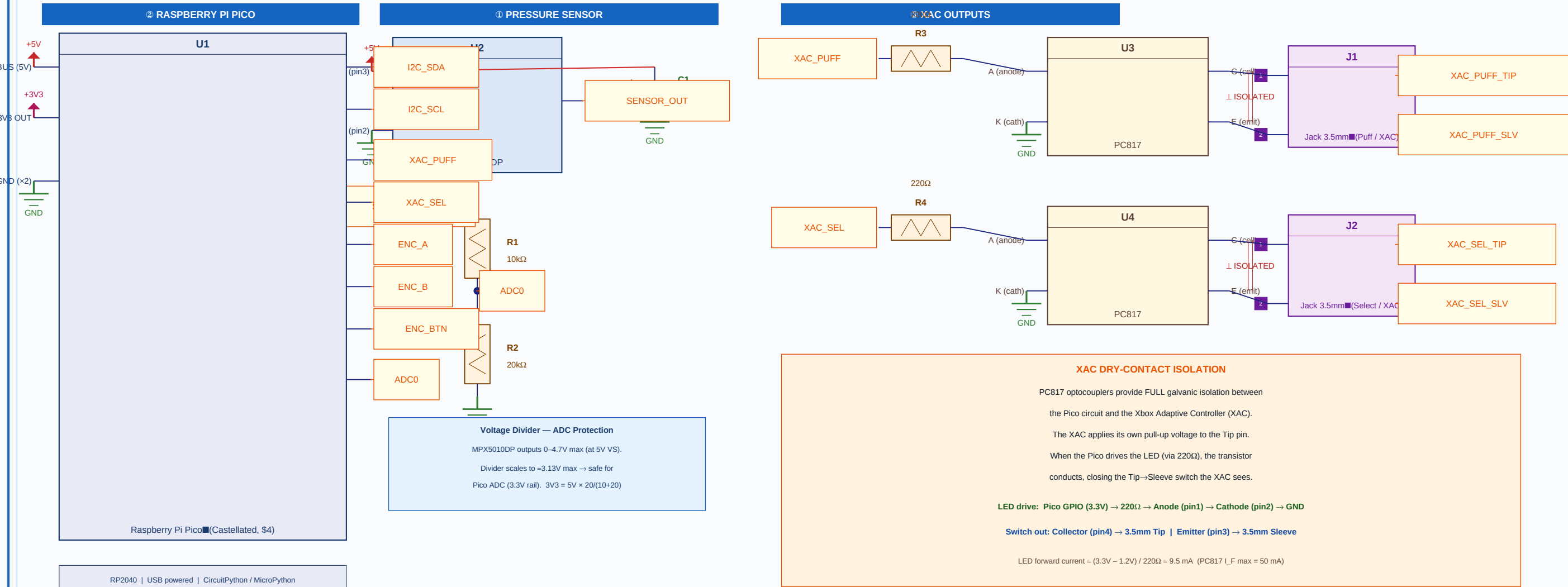
T-REX SIP-N-PUFF HID — SCHEMATIC

Raspberry Pi Pico | MPX5010DP | XAC Optocoupled Outputs | Encoder + I2C Interface

Rev 1.0

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NET LABEL CROSS-REFERENCE	
SENSOR_OUT	MPX5010 VOUT → R1 pin1 (top of voltage divider)
ADC0	R1 pin2 / R2 pin1 junction → Pico GP26 (ADC0)
XAC_PUFF	Pico GP16 → R3 pin1 (left) → PC817 U3 Anode
XAC_SEL	Pico GP17 → R4 pin1 (left) → PC817 U4 Anode
OPTO1_A	R3 pin2 (right) → PC817 U3 Anode (same net as above)
OPTO2_A	R4 pin2 (right) → PC817 U4 Anode (same net as above)
XAC_PUFF_TIP	PC817 U3 Collector → J1 Tip (3.5mm to XAC)
XAC_PUFF_SLV	PC817 U3 Emitter → J1 Sleeve
XAC_SEL_TIP	PC817 U4 Collector → J2 Tip (3.5mm to XAC)
XAC_SEL_SLV	PC817 U4 Emitter → J2 Sleeve
ENC_A	Pico GP18 → J3 pin1 (Rotary Encoder Phase A)
ENC_B	Pico GP19 → J3 pin2 (Rotary Encoder Phase B)
ENC_BTN	Pico GP20 → J3 pin3 (Encoder Push Button)
I2C_SDA	Pico GP4 → J4 pin3 (I2C Data)
I2C_SCL	Pico GP5 → J4 pin4 (I2C Clock)

BILL OF MATERIALS				
Ref	Value	Part / Description	Type	Notes
U1	Raspberry Pi Pico	Castellated pins (\$4 version)	MCU	Pin 40=VBUS, 36=3V3, 38/3=GND, 6=GP4, 7=GP5, 21=GP16, 22=GP17, 24=GP18, 25=GP19, 26=GP20, 31=GP26
U2	MPX5010DP	NXP Unibody SENSOR4F	Pressure Sensor	Pin1=VOUT, Pin2=GND, Pin3=VS(+5V). Pins 4,5,6=NC. Output 0–4.7V @ 5V supply
U3	PC817	DIP-4 Optocoupler	Opto (Puff)	A=pin1, K=pin2, E=pin3, C=pin4. CTR≥50%
U4	PC817	DIP-4 Optocoupler	Opto (Select)	Same as U3
R1	10kΩ ½W 1%	Resistor (SMD 0402 or THT)	Voltage Div	Top of divider: SENSOR_OUT → R1 → ADC0
R2	20kΩ ½W 1%	Resistor (SMD 0402 or THT)	Voltage Div	Bottom of divider: ADC0 → R2 → GND. Scales 4.7V → 3.13V
R3	220Ω ½W	Resistor (SMD 0402 or THT)	LED limit	GP16 → R3 → U3 Anode. Current = 9.5mA